

# Machine Learning-Based Methods for Automatic Compensation of Dead-time Distortion: A Comparative Study

Hideki Ayano<sup>\*a)</sup> Senior Member, Makoto Ohmi<sup>\*</sup> Non-member  
 Yuto Omac<sup>\*\*</sup> Non-member, Yoshihiro Matsui<sup>\*\*\*</sup> Senior Member

(Manuscript received December 23, 2024, revised May 20, 2025)

In many power converters, a compensated voltage reference for dead-time distortion is typically preset during the design stage. However, this reference may deviate from the optimal value when semiconductor devices are replaced or when device characteristics change due to operating conditions. As a result, there is a growing need for active adjustment of the compensated reference. This paper evaluates the effectiveness of machine learning techniques for this task, and compares three approaches: (i) an image classification model, (ii) a waveform feature-based linear regression model, and (iii) a one-dimensional CNN regression model using time-series data. Results show that method (iii) outperforms the others.

**Keywords:** Dead-time compensation, machine learning, auto-tuning, convolutional neural networks (CNNs)

## 1. Introduction

Power converters have been widely used in various applications to save energy. These converters typically have a compensated reference for dead-time distortion that is pre-set during the design phase<sup>(1)</sup>. However, changes in device performance due to operating conditions or deterioration over time may increase the current distortion caused by dead-time. Additionally, when semiconductor devices in a power converter are replaced, a different type may be used. In such cases, it is necessary to actively adjust the new compensated reference. To address this issue, various methods have been proposed<sup>(2)–(6)</sup>. In ref. (2), a method is proposed to automatically adjust the compensated reference for dead-time distortion in a motor drive inverter based on the current distortion component. This method applies a sinusoidal current to the motor windings while the motor is at a standstill. This method calculates the distortion index from the product of the measured current and a third-harmonic signal, and adjusts the compensated reference to eliminate it. In ref. (3), a dead-time compensation method is proposed using a dead time compensation voltage model derived from a circuit model that assumes a virtual capacitor is connected to the semiconductor power device in parallel. The proposed model is based on a parasitic capacitance model, which is able to capture the phe-

nomenon that the dead time voltage becomes smaller as the current becomes smaller due to the switching characteristics of the power element near the current zero crossing. The proposed method includes an automatic adjustment scheme that performs online tuning of parameters based on current ripple information. In ref. (4), an automatic phase error measurement method using a look-up table is proposed, along with an output voltage compensation technique applied at each half of the carrier period, which enables error compensation for each switching event. The proposed compensation method is more effective than conventional methods based on the average error voltage, especially when the current ripple is large and the current is near 0 A. In ref. (5), an adaptive linear neuron-based nonlinearity compensation strategy is proposed, which utilizes only the d-axis current and rotor position angle, without requiring electrical parameters or phase current polarity information. By minimizing the feedback error of the d-axis current, the amplitude of the disturbance compensation voltage is self-tuned using the least mean square algorithm. The proposed method is distinctive in that it enables online estimation of the compensation voltage without the need for complex mathematical computations or additional hardware. In ref. (6), a simple parameter-independent online compensation scheme for dead time and inverter nonlinearity based on waveform analysis is presented for a PWM inverter-fed IPMSM drive. The proposed scheme extracts the voltage distortion caused by dead time and inverter nonlinearity from the effects of parameter variation by monitoring changes in the current slope induced by these nonlinearities. Without requiring any additional hardware, the proposed scheme can effectively compensate for dead time and inverter nonlinearity even in the presence of parameter uncertainty.

In recent years, machine learning technology has been introduced into industrial equipment and home appliances. For example, it can detect anomalies and failures quickly and ac-

This paper is based on Reference (15), which published in IEEJ This paper is based on Reference (16), which published in the International Conference on Electrical Machines and Systems (2024) ©2024 IEEJ.

a) Correspondence to: ayano@tokyo-ct.ac.jp

\* National Institute of Technology, Tokyo College  
 1220-2, Kunugida-machi, Hachioji, Tokyo, Japan 193-0997

\*\* Nihon University  
 1-2-1, Izumi-chou, Narashino, Chiba, Japan 275-8575

\*\*\* Fukuoka Institute of Technology  
 3-30-1, Waziro-higashi, Higashi-ku, Fukuoka, Japan 811-0295

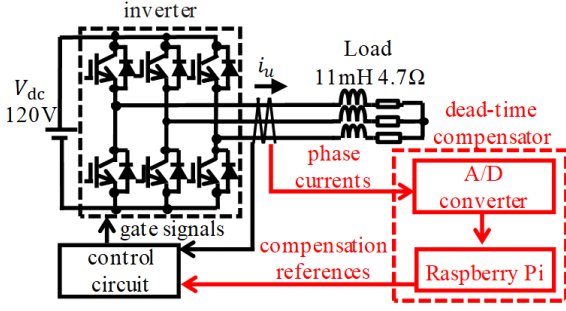


Figure 1 System configuration of power converter.

curately by utilizing features undetectable by humans<sup>(7)–(10)</sup>. Moreover, machine learning can improve productivity and reliability while reducing operating costs<sup>(11)–(14)</sup>. Hence, it has been touted as a solution to various problems.

This paper proposes a technique to estimate the current distortion caused by dead-time and automatically adjust the compensated reference using a new external dead-time compensator<sup>(15)(16)</sup>. A Raspberry Pi, which is a single-board computer with relatively large memory, is employed as the compensator. This enables advanced calculations that are difficult to realize with a general-purpose microcontroller used in inverter drives. This paper evaluates the effectiveness of applying machine learning techniques to the adjustment of the compensated reference. It compares three approaches: (i) an image classification model, (ii) a waveform feature-based linear regression model, and (iii) a one-dimensional CNN (Convolutional Neural Network) regression model with time-series data.

## 2. Configuration of the Evaluation System

Fig. 1 shows the system configuration of the power converter used in the evaluation, which consists of a three-phase inverter with a DC link voltage of 120 V and an RL load ( $R = 4.7 \Omega$ ,  $L = 11 \text{ mH}$ ). This paper deals with a dead-time distortion and evaluates it with the RL load to remove the influence of nonlinear errors such as mechanical losses in the motor. The output frequency is set to 50 Hz. The carrier frequency and dead-time are 10 kHz and  $5 \mu\text{s}$ , respectively. The reference output phase voltage is set to 15 V, corresponding to a modulation ratio of 0.12, which is considered low enough for the effects of current distortion to be significant. In order to evaluate current distortion caused solely by dead time, the inverter is driven in open loop without current control. This approach suppresses waveform distortion caused by the control gain characteristics of current feedback control, enabling the evaluation of effects attributable only to dead time.

In Fig. 1, a dead-time compensator is installed to independently calculate the output current characteristics apart from the control circuit of the inverter. The Raspberry Pi in the compensator stores the current waveforms acquired by the AD converter in the memory and performs machine learning calculations using the current information. The results calculated in the compensator are transmitted to the control circuit via serial communication and used to correct the compensated reference for the dead-time.

Since the turn-off time of switching devices such as IGBTs is generally longer than the turn-on time, a dead-time is

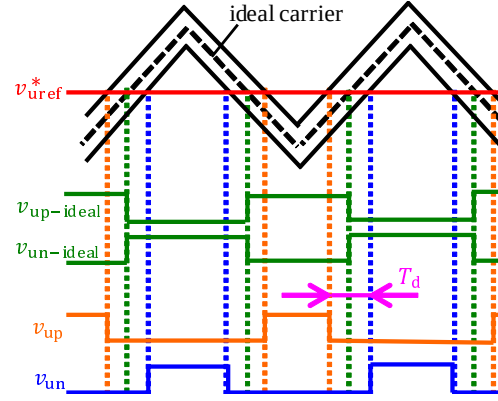


Figure 2 PWM references with the dead time.

required to prevent a short-circuit fault where the switching devices in the upper and lower arms are turned on simultaneously. Fig. 2 shows an example of PWM references with the dead-time. The dotted triangular wave represents the carrier under ideal conditions with no dead-time. When the dead-time is provided, the solid-line carriers are compared with the voltage reference  $v_{uref}^*$ . As a result, the references  $v_{up}$  and  $v_{un}$  for driving the upper and lower arm switching devices have the dead-time  $T_d$ . On the other hand, during  $T_d$ , the output values of the phase voltages become either 0 V or  $V_{dc}$  [V], depending on the direction of the phase currents flowing into the load. Consequently, the pulse widths of  $v_{up}$  and  $v_{un}$  differ from those of the ideal PWM pulses ( $v_{up-ideal}$ ,  $v_{un-ideal}$ ) obtained by comparing the ideal triangular wave carrier with  $v_{uref}^*$ . This results in a current distortion, which is especially noticeable at low modulation rates. To compensate for this current distortion, a method is generally employed that detects the direction of the phase current and adds or subtracts a preset compensation voltage established during the design stage to or from  $v_{uref}^*$ .

Fig. 3 shows the simulation results of the phase currents when the compensated reference for dead-time  $c_{dt}$  is varied. Altair PSIM was used for the simulations. An ideal switch is used as the switching device. In this paper, the appropriate value of  $c_{dt}$  is normalized to 1.0 and treated as a constant over one period of the fundamental component. When  $c_{dt}$  is 0 (no compensation), the phase current waveform contains distortion. On the other hand, when  $c_{dt}$  is 1.0 (with appropriate compensation), the distortion is reduced and the current waveform becomes sinusoidal. When  $c_{dt}$  is 2.0 (overcompensation), the phase current also contains distortion. In this case, the shape of the current distortion is different from that of  $c_{dt} = 0$ . In addition, since the amplitude of the current is not controlled, the voltage reference pulse width extends, and the amplitude of the phase currents increases with increasing  $c_{dt}$ . Note that this paper does not consider distortion that occurs when the current is extremely small due to the nonlinear characteristics of actual switching devices. This makes it difficult to completely suppress distortion as observed in simulation results. Under such conditions,  $c_{dt}$  is selected so that the waveform approximates the low-distortion waveform obtained through machine learning, as described in Chapters 3 to 5.

Figs. 4 and 5 show the experimental results of the phase

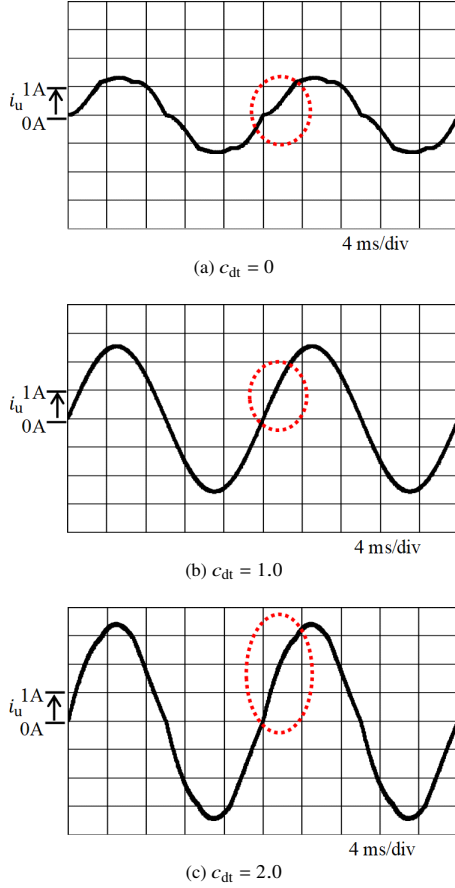


Figure 3 Simulation results of the phase currents.

currents and their FFT analyses. Figs. 4(a) and 5(a) show the characteristics without compensation ( $c_{dt} = 0$ ), Figs. 4(b) and 5(b) show the characteristics when the compensation is adjusted to an appropriate compensation ( $c_{dt} = 1.0$ ), and Figs. 4(c) and 5(c) show the characteristics with overcompensation ( $c_{dt} = 2.0$ ). The compensation value for  $c_{dt} = 1.0$  was initially estimated based on the dead-time duration and the rise/fall characteristics obtained from the device datasheet. This was followed by fine adjustments to determine the true  $c_{dt}$  that minimizes harmonic distortion in the FFT analysis. The derived value was also used in Figs. 9, 11, and 15 in later chapters. As shown in Figs. 4 and 5, the harmonic components and waveform distortions increase when  $c_{dt} = 0, 2.0$ . On the other hand, when  $c_{dt} = 1.0$ , the harmonic components are reduced and waveform distortion becomes smaller, as in the simulation<sup>†</sup>. Compared to those with  $c_{dt} = 0$ , the 5th and 7th harmonic components are reduced to 1/6 and 1/5, respectively.

<sup>†</sup> In this evaluation, as described in Chapters 3 to 5, the compensated reference  $c_{dt}$  is updated periodically at each fundamental period based on waveforms acquired over five cycles. The waveforms shown in Figs. 4(a) (c) are corrected, resulting in the compensated waveform presented in Fig. 4(b). In practice, the value of  $c_{dt}$  exhibits slight variations with each update, which causes minor fluctuations in the waveform. Nevertheless, the overall waveform remains close to that shown in Fig. 4(b).

### 3. Computation of Dead-time compensation using Image Classification Model

**3.1 System Configuration of Image Classification Model** This chapter evaluates a method for predicting compensated references for dead-time using machine learning with an image classification model. Fig. 6 shows a schematic diagram of an automatic adjustment system using this model. The dead-time compensator performs data preprocessing on the acquired waveform information and generates image data. The data is fed into the image classification model to predict the compensated reference. Using waveform data over five cycles obtained from the A/D converter, the compensator averages the individual prediction results from each cycle to determine the final compensated reference. Subsequently, the compensator calculates a command to apply the appropriate compensation to the reference signal and sends it to the control circuit. The control circuit updates the compensated reference based on the command from the compensator and controls the converter system.

First, the procedure for creating the image classification model is described. “Teachable Machine,” which is a web-based tool provided by Google, is used to train the models. It uses transfer learning with MobileNet, one of the convolutional neural networks for solving the 1000-class classification problem defined in ILSVRC. This makes it possible to create a model that solves the image classification problem. The model evaluated in this paper predicts classes with  $c_{dt}$  between 0 and 2.0 in increments of 0.1. Therefore, it is necessary to collect image data of one-period waveforms as  $c_{dt}$  is varied.

Fig. 7 shows examples of the waveform images used for training. Five cycles of waveform data are detected by the A/D converter in the compensator. To generalize the evaluation regardless of load conditions (for example, load impedance), the 5-cycle current waveform data are scaled (normalized) after removing skipped values and filtering noise components. The minimum and maximum values are then set to  $-1$  and  $1$ , respectively, under all conditions. This increases the difficulty of prediction. The waveform data are divided into 1-cycle data. The sampling frequency for waveform detection by the A/D converter is 100 kHz, resulting in 2000 data points per cycle. 200 image data samples are collected for each increment of  $c_{dt}$  of 0.1. Then, 150 samples from each are used for training, totaling 3150 samples. The remaining 50 samples from each, totaling 1050 samples, are used for validation. The data are converted to  $224 \times 224$  pixel PNG files by Matplotlib and used as training data for the machine learning model. These are carried out in the pre-processing and image generation stage shown in Fig. 6.

Fig. 8 shows the transitions of accuracy and loss during the training process of the image classification model on the Teachable Machine. It is confirmed that both the training and validation data converge under the condition that the epochs are 200. Therefore, this condition is used for the evaluation.

**3.2 Evaluation of the Method using Image Classification Model** Fig. 9 shows the prediction results of the compensated reference by the image classification model using evaluation data with known true values. The top, center, and bottom lines of the error bars represent the maximum, mean, and minimum values, respectively.

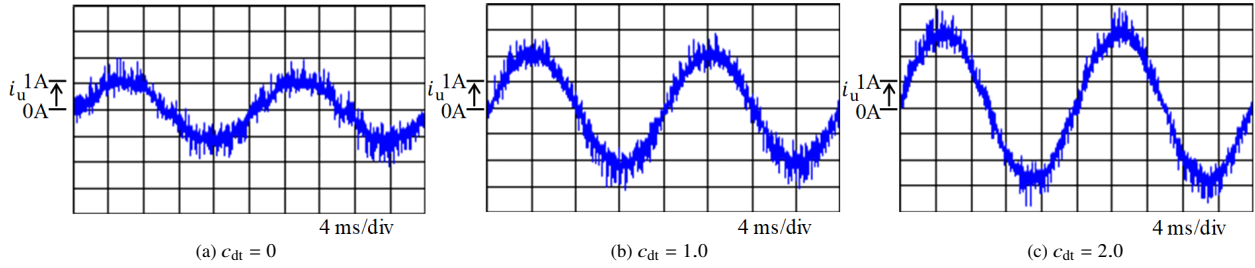


Figure 4 Experimental results of the phase currents.

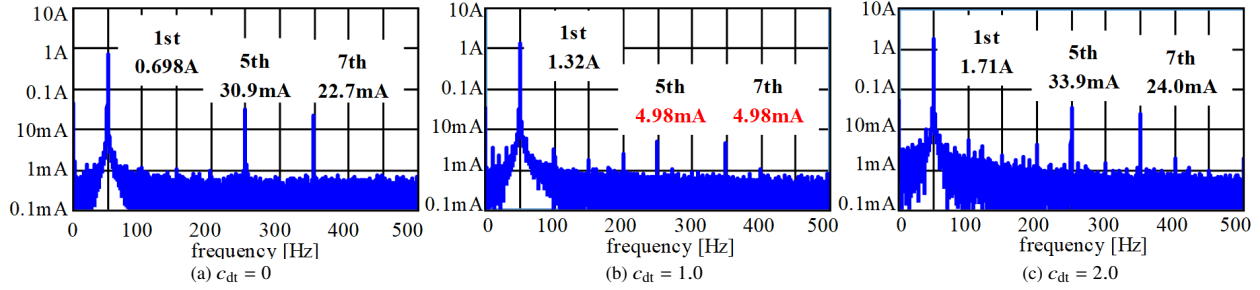


Figure 5 FFT analyses of the phase currents.

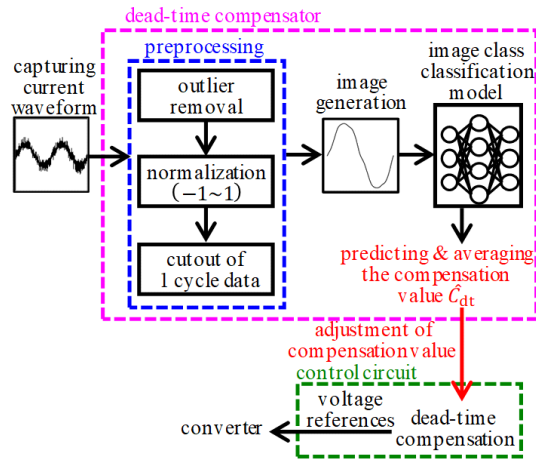


Figure 6 Schematic diagram of an automatic adjustment system using image classification model.

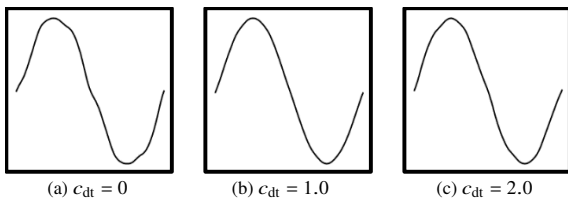


Figure 7 Examples of the waveform images used for training.

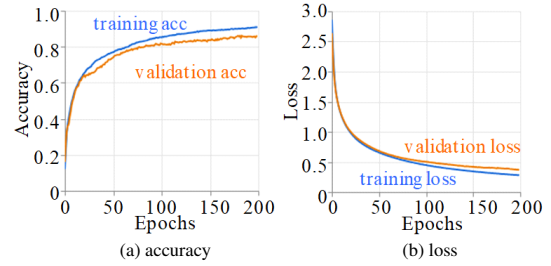


Figure 8 The transitions of accuracy and loss during the training process of the image classification model.

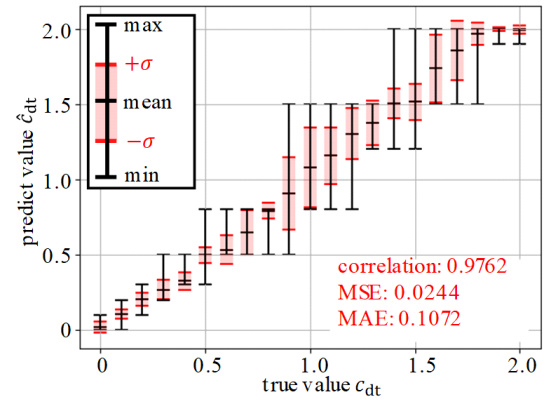


Figure 9 Prediction results of the compensated reference by the image classification model.

mean, and minimum predicted values, respectively. The upper and lower edges of the boxes indicate the standard deviations,  $+\sigma$  and  $-\sigma$ . The correlation coefficient between the predicted and true values, mean squared error (MSE) and mean absolute error (MAE) of the difference between the predicted and true value are 0.9762, 0.0244, and 0.1072, respectively. The correlation coefficient is a statistical measure that indicates the degree of correlation between two variables. Its value ranges from  $-1.0$  to  $1.0$ , with values closer to  $-1.0$  indicating a negative correlation, values closer to  $1.0$  indi-

cating a positive correlation, and values near 0 indicating no correlation. MSE and MAE indicate higher accuracy when their values are smaller. In Fig. 9, each standard deviation  $\pm\sigma$  indicates an accuracy of approximately  $\pm 0.2$  relative to the true value for  $c_{dt}$  less than 0.8. However, for  $c_{dt}$  greater than 0.9, the  $\pm\sigma$ s of the predicted values become large, and the mean of the predicted value may differ from the true value. When  $c_{dt}$  exceeds 1.8, the estimation results are misjudged as 2.0. In other words, when  $c_{dt}$  is large, this method has difficulty distinguishing between waveforms, making accurate

prediction difficult.

The decrease in prediction accuracy with this model may be attributed to the process of converting the current waveform data into a  $224 \times 224$  pixel image. This conversion generates a large number of white pixels that are unrelated to the actual signal, leading to the inclusion of irrelevant information for prediction. Consequently, this increases the learning difficulty of the model, potentially causing a decline in prediction performance. Another possible factor is that the relatively coarse resolution of  $224 \times 224$  pixels may make it difficult to capture fine features of the signal. In addition, this model assigns prediction labels at 0.1 intervals from 0 to 2.0 and evaluates correctness strictly for each label. As a result, even slight deviations are regarded as errors, complicating the training process for the model.

In this model, the time required to determine  $c_{dt}$  is approximately 10 seconds. The  $c_{dt}$  is continuously adjusted and updated at 10-second intervals. Even when the adjustment has been started from under-compensation or over-compensation conditions as shown in Figs. 4(a) and 4(c), it has been confirmed that the compensated references for dead-time converge to a near-optimal value after several update cycles, and the current waveform has been automatically adjusted as shown in Fig. 4(b).

#### 4. Computation of Dead-time Compensation using Waveform Feature-based Linear Regression Model

**4.1 System Configuration of Waveform Feature-based Linear Regression Model** This chapter evaluates a method for predicting compensated references for dead-time using a linear regression model learned from features extracted out of waveform time-series data. In learning and evaluating the machine learning model, the same training and evaluation data for current waveforms as in the previous chapter are used. Specifically, 200 waveform data samples are collected for each increment of  $c_{dt}$  of 0.1. Then, 150 samples from each are used for training, totaling 3150 samples. The remaining 50 samples from each, totaling 1050 samples, are used for validation.

Fig. 10 shows a schematic diagram of the automatic adjustment system using the waveform feature-based linear regression model. The difference from the processing in the previous method is that the image data generation part and the image classification model part are changed to a feature-based linear regression model. In addition, similar to the image classification model, the final prediction result is obtained by averaging the prediction results from five cycles of waveform data. This improves overall prediction accuracy. This method has the advantages of smaller model size and faster prediction compared to the image classification model. Furthermore, evaluating the weights of the learned model allows consideration of the features that are important for predictions.

Next, the method for generating a feature-based linear regression model is explained. The first and second columns of Table 1 show the used features. In this evaluation, 18 features were used, which were selected based on a preliminary evaluation as those expected to exhibit the relevant characteristics. The extracted feature data are normalized (0 ~ 1.0).

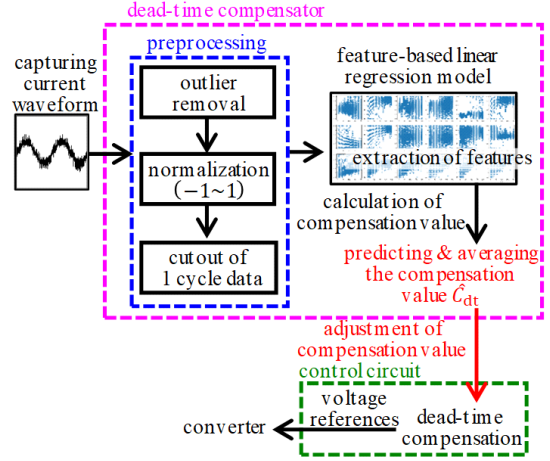


Figure 10 Schematic diagram of an automatic adjustment system using waveform feature-based linear regression model.

Table 1 Features and weights of the waveform feature-based linear regression model.

|          | Feature                       | Weight $w^*$ |
|----------|-------------------------------|--------------|
| $a_1$    | Mean                          | -0.0169      |
| $a_2$    | Variance                      | 3.4183       |
| $a_3$    | Skew                          | -0.0364      |
| $a_4$    | Kurtosis                      | -0.0759      |
| $a_5$    | Median                        | -0.0215      |
| $a_6$    | Sum of absolute               | -2.9234      |
| $a_7$    | Min                           | -0.0195      |
| $a_8$    | Max                           | 0.0296       |
| $a_9$    | Index of the minimum          | 0.0066       |
| $a_{10}$ | Index of the maximum          | -0.0024      |
| $a_{11}$ | Samples in range -1.0 to -0.9 | 0.034        |
| $a_{12}$ | Samples in range -0.9 to -0.8 | 0.2855       |
| $a_{13}$ | Samples in range -0.8 to -0.7 | 0.2533       |
| $a_{14}$ | Samples in range -0.7 to -0.6 | -0.0018      |
| $a_{15}$ | Samples in range 0.0 to 0.1   | 0.1786       |
| $a_{16}$ | Samples in range 0.1 to 0.2   | 0.3596       |
| $a_{17}$ | Samples in range 0.2 to 0.3   | 0.3228       |
| $a_{18}$ | Samples in range 0.3 to 0.4   | 0.2083       |

The ElasticNet class in Scikit-learn is used for learning for the weights. In this evaluation, in addition to general metrics such as Mean, Variance, and Skew, features were selected to achieve favorable values for the correlation coefficient, MSE, and MAE. In particular, features  $a_{11}$  to  $a_{18}$  were defined by specifying amplitude ranges and counting the number of data points within each range, based on the characteristics of waveform distortion, such as bulging and denting, observed during under- and over-compensation, as confirmed through simulations. The waveform data obtained from the A/D converter over five cycles were normalized so that the maximum and minimum values were set to 1 and -1, respectively. As a result, each waveform is represented as time-series data with values ranging from -1 to 1. Accordingly, the number of data points within a specific amplitude range (e.g., from  $x$  to  $y$ ) is used as a feature ("Samples in range  $x$  to  $y$ "). An evaluation using all amplitude segments divided into 0.1 intervals from -1 to 1 was also conducted. However, little improvement in prediction accuracy was observed. Therefore, features  $a_{11}$  to  $a_{18}$  were selected as the final feature set. While it is possible that the evaluation accuracy might be further improved by increasing the number of features, the initial evaluation in this

study was conducted using the 18 features in Table 1, which are considered optimal at this stage.

The predictive model for linear regression using Elastic Net is defined by the inner product of the feature vector  $\mathbf{a}$  and the weight vector  $\mathbf{w}$ , as follows:

$$\hat{c}_{dt} = \mathbf{w}^\top \mathbf{a}. \dots\dots\dots (1)$$

Here,  $\mathbf{w} = [w_0 \dots w_d]^\top$ ,  $\mathbf{a} = [a_0 \dots a_d]^\top$ . In this paper, 18 features defined in Table 1 are used, resulting in  $d = 18$ . Specifically,  $a_1, \dots, a_{18}$  represent the 18 features defined in Table 1, and  $w_1, \dots, w_{18}$  are their corresponding weights. Additionally,  $a_0 = 1$ , making  $w_0$  the intercept term. The objective function for determining the weight vector  $\mathbf{w}$  is given by:

$$L^*(\mathbf{w} \mid \alpha, \beta, \gamma) = \alpha E(\mathbf{w}) + \beta \|\mathbf{w}\|_1 + \gamma \|\mathbf{w}\|_2^2, \dots\dots (2)$$

where  $E(\mathbf{w})$  represents the mean squared error (MSE) calculated using the training data,  $\|\mathbf{w}\|_1$  is the  $L_1$  norm, and  $\|\mathbf{w}\|_2^2$  is the  $L_2$  norm. The first, second and third terms contribute to fitting to the training data, reducing the weights of unnecessary features to zero and suppressing overfitting by reducing the weights, respectively. Using the training data, the optimal weight vector  $\mathbf{w}^*$  is determined by solving the following optimization problem:

$$\mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^d}{\operatorname{argmin}} L^*(\mathbf{w} \mid \alpha, \beta, \gamma). \dots\dots\dots (3)$$

To evaluate how well the obtained weight vector  $\mathbf{w}^*$  fits to new data, the mean absolute error (MAE)  $L^\dagger(\mathbf{w}^*)$  is calculated using validation data, which is independent of the training data. A lower MAE  $L^\dagger(\mathbf{w}^*)$  indicates that  $\mathbf{w}^*$  provides better accuracy on the validation data. However, the selected  $\mathbf{w}^*$  depends on the parameters  $\alpha, \beta, \gamma$ , which are constants in Eq. (3). To achieve a weight vector  $\mathbf{w}^*$  that best fits new data, the optimal hyperparameters  $\alpha, \beta, \gamma$  are determined as follows:

$$(\alpha^\dagger, \beta^\dagger, \gamma^\dagger) = \underset{\alpha, \beta, \gamma}{\operatorname{argmin}} L^\dagger(\mathbf{w}^*). \dots\dots\dots (4)$$

Consequently, the optimal weight vector obtained using the optimal hyperparameters  $\alpha^\dagger, \beta^\dagger, \gamma^\dagger$  is:

$$\mathbf{w}^\dagger = \underset{\mathbf{w} \in \mathbb{R}^d}{\operatorname{argmin}} L^*(\mathbf{w} \mid \alpha^\dagger, \beta^\dagger, \gamma^\dagger), \dots\dots\dots (5)$$

which is used as the final linear regression model. Eqs. (3) ~ (5) do not guarantee exact optimal solutions. Therefore, approximate solutions were obtained: coordinate descent (1000 iterations) was employed for Eqs. (3) and (5), while Tree-structured Parzen Estimators (TPE, 250 iterations) were used for Eq. (4). The values of  $\alpha, \beta, \gamma$  were determined via 3-fold cross-validation (3-fold CV). The hyperparameters obtained through this process were  $\alpha^\dagger = 2.2778 \times 10^{-2}$ ,  $\beta^\dagger = 2.942 \times 10^{-11}$ , and  $\gamma^\dagger = 5.014 \times 10^{-8}$ . The resulting weight vector  $\mathbf{w}^\dagger$  is listed in third column of Table 1. The intercept  $w_0$  is set to  $-0.642$ . Since the features  $a_i$  are normalized, the larger the absolute value of the weight, the greater its impact on the prediction results.

#### 4.2 Evaluation of the Method using Waveform Feature-based Linear Regression Model

Fig. 11 shows

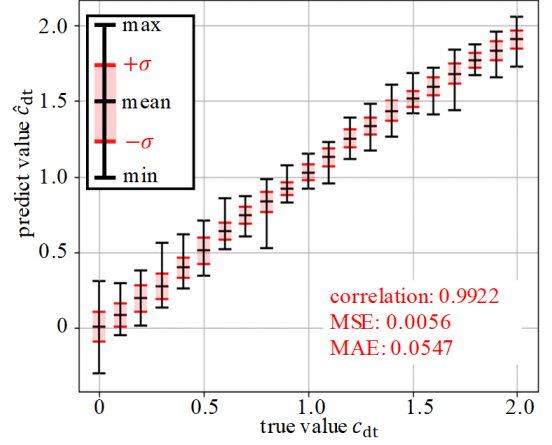


Figure 11 Prediction results of the compensated reference by waveform feature-based linear regression model.

the prediction results of the compensated reference by the waveform feature-based linear regression model using evaluation data with known true values. In each  $c_{dt}$ , the difference between the minimum and maximum values is small, and each standard deviation ( $\pm\sigma$ ) confirms that the accuracy is within  $\pm 0.1$  of the true value. Furthermore, the correlation, MSE, and MAE are 0.9922, 0.0056, and 0.0547, respectively, indicating that the estimation accuracy of the compensated reference is higher than that of the image classification method in chapter 3. This improved prediction performance can be attributed to the fact that the model does not convert current waveform data into images but instead learns directly using the mean squared error. As a result, issues such as the inclusion of unnecessary information through image conversion, the difficulty of feature extraction caused by resolution constraints, and the increased learning difficulty caused by strict label matching are mitigated.

In this machine learning model, the determination of  $c_{dt}$  requires approximately 10 seconds. The  $c_{dt}$  is updated in each cycle, and once it converges to the appropriate value, the system operates with only minimal fluctuations around it. By using this model, it has been confirmed that even when the adjustment is started from under-compensation or over-compensation conditions as shown in Figs. 4(a) and 4(c), the compensated references for dead-time can be adjusted close to the appropriate value in almost a single adjustment, and the current waveform is automatically adjusted to that shown in Fig. 4(b). In conclusion, the effectiveness of the automatic adjustment method for dead-time compensation using the linear regression model, which extracts features from waveform the time-series numerical data of the waveforms, has been confirmed.

### 5. Computation of Dead-time Compensation using One-dimensional CNN Regression Model with Time Series Data

**5.1 System Configuration of One-Dimensional CNN Regression Model with Time Series Data** This chapter evaluates a method for predicting compensated references for dead-time using a one-dimensional CNN regression model learned from time-series data of waveforms. In learning and evaluating the machine learning model, the same training and

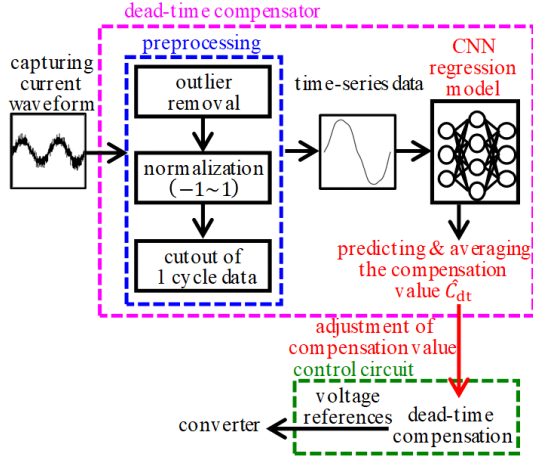


Figure 12 Schematic diagram of an automatic adjustment system using one-dimensional CNN regression model.

evaluation data for current waveform data as used in the previous chapters are employed. Specifically, 200 waveform data samples are collected for each increment of  $c_{dt}$  of 0.1. Then, 150 samples from each are used for training, totaling 3150 samples. The remaining 50 samples from each, totaling 1050 samples, are used for validation.

Fig. 12 shows a schematic diagram of the automatic adjustment system using the one-dimensional CNN regression model. The difference between the processing of this method and the previous methods is the use of a CNN regression model as the machine learning model for estimating the compensated references for dead-time. As with the previous two methods, this method uses waveform data over five cycles and averages the prediction results from each cycle to obtain the final prediction result. This improves the accuracy of the prediction results. This method, which can directly use current waveform data for learning, has advantages in ease of implementation and high scalability compared to the machine learning models discussed in the previous chapters.

The following explains how to create a CNN regression model trained with time-series data. The time-series data of current waveforms are preprocessed by outlier processing, normalization ( $-1.0 \sim 1.0$ ) and one-cycle data cutout. Then, the preprocessed one-cycle waveform time-series data are used as the training data. Each data is one-dimensional data with 2000 data points. Fig. 13 shows the structure of the CNN model. The input is a one-dimensional array of shape (2000, 1). It passes through an input layer, convolutional layers, max pooling layers, a dense layer, a flattening layer, an output layer, resulting in a scalar output. Keras, a neural network library developed by Google, was used to create and train the model. Mean squared error (MSE) was used as the loss function and Adaptive Moment Estimation (Adam) was used as the optimization algorithm. Fig. 14 shows the loss in the training process of the CNN model. It can be confirmed that the loss for both the training and validation data converges after about 20 epochs. The batch size and epochs for training were set to 32 and 100, respectively.

## 5.2 Evaluation of the Method using One-Dimensional CNN Regression Model with Time Series Data

Fig. 15

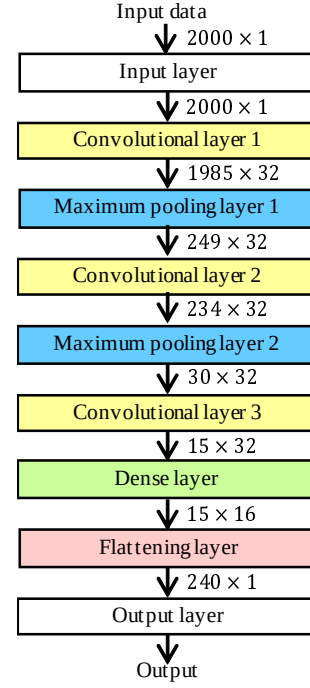


Figure 13 Structure of the CNN model.

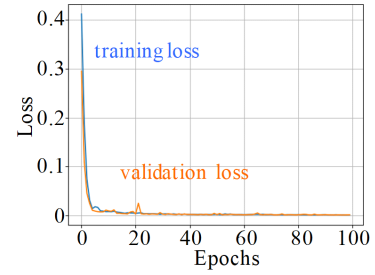


Figure 14 Loss in the training process of the CNN model.

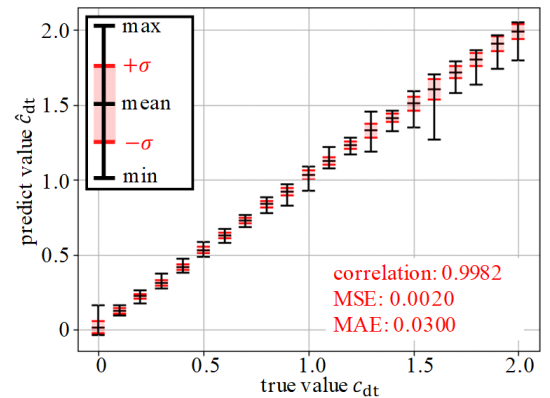


Figure 15 Prediction results of the compensated reference by the CNN regression model.

shows the prediction results of the compensated reference by the CNN regression model using evaluation data with known true values. The difference between the minimum and maximum values is small, and each standard deviation ( $\pm\sigma$ ) confirms that the accuracy is within  $\pm 0.1$  of the true value. The correlation, MSE, and MAE are 0.9982, 0.0020, and 0.0300, respectively, indicating that the estimation accuracy of the

compensated reference is higher than those of the methods in the previous chapters. One factor that may explain why this model demonstrates higher performance than the waveform feature-based linear regression model is the difference in constraints on the functional form. The waveform feature-based linear regression model assumes a proportional relationship between input and output, whereas this model does not impose such a constraint. Therefore, if nonlinear relationships, such as non-monotonic, logarithmic, or exponential relationships, exist between input and output, this model may be better suited to capture such complexity. Another possible factor is the selection of features. In the waveform feature-based linear regression model, the features considered to be effective were used. However, it is possible that some unnecessary features for prediction were included, and that more effective features may have existed among those not selected in this study. In contrast, this model can automatically learn effective features that reduce the mean squared error of the training data. Therefore, given sufficient training data, this model may also be able to extract effective features for new data.

In this machine learning model, the determination of  $c_{dt}$  requires approximately 10 seconds. The  $c_{dt}$  is updated in each cycle, and once it converges to the appropriate value, the system operates with only minimal fluctuations around it. By using this model, it has been confirmed that even when the adjustment is started from under-compensation or over-compensation conditions as shown in Figs. 4(a) and 4(c), the compensated references for dead-time can be adjusted close to the appropriate value in almost a single adjustment, and the current waveform is automatically adjusted to that shown in Fig. 4(b).

In this paper, the modulation rate and output frequency are evaluated under constant conditions, but it is expected that the magnitude of distortion will change when the conditions are changed. By using the CNN model to learn waveform information each time, it is believed that the compensated reference corresponding to those conditions can be estimated.

## 6. Conclusions

This paper has proposed a technique to estimate the current distortion caused by dead-time and automatically adjust the compensated reference using a new external dead-time compensator. In addition, the effectiveness of applying machine learning techniques to the adjustment of the compensated reference has been evaluated by comparing three approaches using current waveform data acquired by an A/D converter: (i) an image classification model based on 224×224 pixel PNG images, (ii) a waveform feature-based linear regression model using 18 waveform features, and (iii) a one-dimensional CNN (Convolutional Neural Network) regression model with time series-data. The following results have been obtained.

- The method using the CNN regression model is more effective than the other methods in terms of correlation, MSE, and MAE.
- The CNN regression model for predicting the compensated reference has been confirmed to have an accuracy within  $\pm 0.1$  standard deviation of the true value.
- The effectiveness of the automatic adjustment method for dead-time compensation using the CNN regression model with the time-series numerical data of the wave-

forms has been confirmed.

In this paper, to evaluate the current distortion caused solely by the dead time itself, the system has been driven in open-loop mode without current control, so that the influence of the feedback gain has been intentionally excluded. The applicability of current feedback to motor drives and other practical applications will be evaluated in future work.

## Acknowledgment

This work was partly supported by JSPS KAKENHI Grant Number JP23K03829.

## References

- (1) T. Mannen, H. Fujita, "Dead-time compensation method based on current ripple estimation," *IEEE Transactions on Power Electronics*, vol. 30, no. 7, pp.4016-4024, 2015.
- (2) D. Li, Y. Funahashi, J. Takagi, "Auto-Tuning Method of Dead-Time Compensation Value for Motor Drive Inverters," *IEEJ Industry Applications Society Conference*, 3-65, 2023 (in Japanese).
- (3) S. Furutani, A. Satake, T. Hozuki, "Inverter Dead Time Compensation Method using On-Line tuning," *IEEJ Journal of Industry Applications*, Vol. 10, No. 2, pp. 264-272, 2021.
- (4) T. Daido and Y. Hayashi, "Output Voltage Error Compensation for Every Half of a Carrier Period in a Voltage Source Inverter," *IEEJ J. Industry Applications*, vol. 8, no. 1, pp. 41-50, 2019.
- (5) Y. C. Pai, J. P. Chang, M. Y. Cheng and T. J. Liang, "Dead-time effects compensation for PMSM drives - an adaptive linear neuron approach," *IEEE International Future Energy Electronics Conference and ECCE Asia (IFEEC 2017)*, 2017.
- (6) D. M. Park and K. H. Kim, "Parameter-independent online compensation scheme for dead-time and inverter nonlinearity in IPMSM drive through waveform analysis," *IEEE Trans. Ind. Elec.*, vol. 61, no. 2, pp. 701-707, 2014.
- (7) D. P. Apsari, J. Jung, D. C. Lee, "Open-Circuit Fault Diagnosis Based on Deep Learning for Four-Level Active Neutral-Point Clamped Inverters," *International Conference on Power Electronics and ECCE Asia (ICPE 2023 - ECCE Asia)*, 2023.
- (8) A. Samanta and S. Williamson, "Machine Learning-based Remaining Useful Life Prediction Techniques for Lithium-ion Battery Management Systems: A Comprehensive Review," *IEEJ Journal of Industry Applications*, Volume 12, Issue 4, pp. 563-574, 2023.
- (9) N. C. Yang, H. Ismail, "Robust Intelligent Learning Algorithm Using Random Forest and Modified-Independent Component Analysis for PV Fault Detection: In Case of Imbalanced Data," *IEEE Access*, Vol. 10, pp. 41119-41130, 2022.
- (10) H. L. Dang, S. Kwak, S. Choi, "Different Domains Based Machine and Deep Learning Diagnosis for DC Series Arc Failure," *IEEE Access*, Vol. 9, pp. 166249-166261, 2021.
- (11) D. P. Apsari, D. C. Lee, "Capacitance and ESR Estimation of DC-link Capacitors in AC Machine Drives Based on Hybrid CNN-Attention Model," *IEEE International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia)*, 2024.
- (12) Y. Wu, K. Shigematsu, Y. Omoto, Y. Ikushima, J. Imaoka, and M. Yamamoto, "Accurate Leakage Inductance Modeling Using an Artificial Neural Network Based on the Dowell Model," *IEEJ Journal of Industry Applications*, Volume 12, Issue 3, pp. 334-344, 2023.
- (13) M. Ziadia, S. Kelouwani, A. Amamou, K. Agbossou, "An Adaptive Regenerative Braking Strategy Design Based on Naturalistic Regeneration Performance for Intelligent Vehicles," *IEEE Access*, Vol. 11, pp. 99573-99588, 2023.
- (14) V. Sudha, K. Vijayarekha, R. K. Sidharthan, N. Prabakaran, "Combined Optimizer for Automatic Design of Machine Learning-Based Fault Classifier for Multilevel Inverters," *IEEE Access*, Vol. 10, pp. 121096-121108, 2022.
- (15) M. Ohmi, H. Ayano, Y. Omae and Y. Matsui, "Automatic Adjustment Technique of Dead-time Compensation in Power Converters with an External Calculation Circuit," *The paper of joint technical meeting on "Semiconductor power converter", "Home and consumer appliances", "vehicle technology", IEE Japan*, SPC-23-252, HCA-23-042, VT-23-065, 2023 (in Japanese).
- (16) H. Ayano, M. Ohmi, Y. Omae and Y. Matsui, "Evaluation of Automatic Adjustment Methods for Voltage Reference to Compensate Dead-time Distortion using Machine Learning," *The 27th International Conference on Electrical Machines and Systems (ICEMS2024-Fukuoka)*, 2024.

**Hideki Ayano** (Senior Member) received the B.S. and M.S. degrees in electrical and electronic engineering from Okayama University, Okayama, Japan, in 1995 and 1997, respectively. In 1997, He was with Hitachi, Ltd. He received Dr. Eng. degree in electrical engineering from Utsunomiya University, Tochigi, Japan in 2006. Since 2011, he has been with the National Institute of Technology, Tokyo College, where he is currently a professor. His research interests include power converters, motor drive, electromagnetic interference problems, etc. He is a Member of the IEEE and a Senior Member of the IEEJ.

**Makoto Ohmi** (Non-member) received the associate degree in electrical engineering and B.S. degree in advanced course of Electrical and Electronic Engineering from National Institute of Technology, Tokyo College, Tokyo, Japan in 2022 and 2024, respectively. He is currently with NTTME CORPORATION. His research interest was power converters.

**Yuto Omae** (Non-member) received the Doctoral degree in engineering from the Department of Information Science and Control Engineering, Nagaoka University of Technology, Japan, in 2016. He is a Lecturer with the Department of Industrial Engineering and Management and the vice director with the Artificial Intelligence Research Center, College of Industrial Technology, Nihon University. His research interests include intelligent informatics, machine learning, and mathematical model. He is a member of the IEEE and IEICE.

**Yoshihiro Matsui** (Senior Member) received the B.E., the M.E. and the D.E. degrees in electronic engineering from the University of Electro-Communications, Tokyo, Japan, in 1985, 1987 and 1999, respectively. From 1987 to 1993, he was with Matsushita Electric Industrial Co., Ltd. He is currently a professor at Fukuoka Institute of Technology, Fukuoka, Japan. His research interests include motion control, system identification, power electronics, etc. He is a member of the SICE and ISCIE, and a Senior Member of the IEEJ.